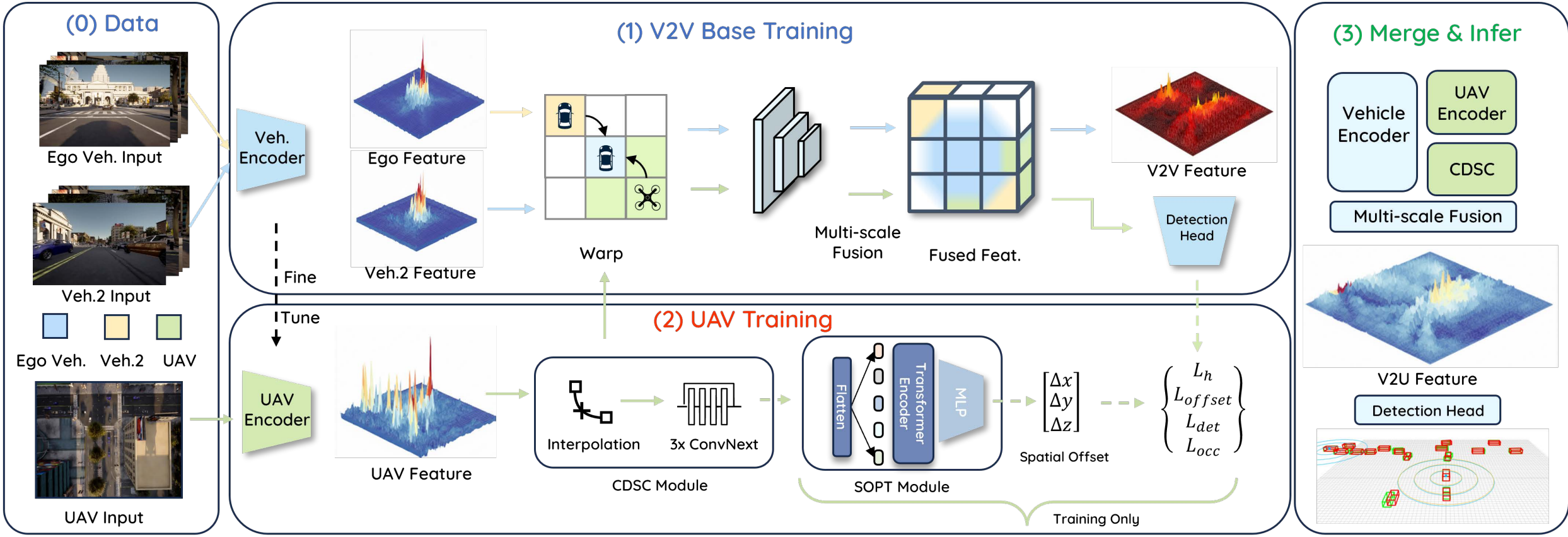
(0) Data
Ego Veh. Input
Veh.2 Input
Ego Veh.
Veh.2
UAV
UAV Input
(1) V2V Base Training
Veh. Encoder
Ego Feature
Veh.2 Feature
Warp
Multi-scale Fusion
Fused Feat.
V2V Feature
Detection Head
Fine
Tune
(2) UAV Training
UAV Encoder
UAV Feature
Interpolation
3x ConvNext
CDSC Module
Flatten
Transformer Encoder
MLP
SOPT Module
Spatial Offset
Training Only
(3) Merge & Infer
Vehicle Encoder
UAV Encoder
CDSC
Multi-scale Fusion
V2U Feature
Detection Head